

Sasi Kanduri

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EDUCATION

Master of Science, Computer Science
Bachelor of Science, Information Technology

California State University, Sacramento
Jawaharlal Nehru Technological University, India

Aug 2022 – Dec 2024
Aug 2013 – May 2017

TECHNICAL SKILLS

Languages/Frameworks: Python, Java, JavaScript, TypeScript, React, Flask, FastAPI, SpringBoot, Swift, LangGraph

Technologies/Software: Git, Maven, UV, Docker, AWS, Linux, Pega

WORK EXPERIENCE

Full Stack Engineer - Office of Water Programs | California, USA

Jun 2023 – Present

- Led the design and development of a state-wide water quality analytics and training platform; Built frontend with React/TypeScript and backend APIs and microservices with Python (Flask/FastAPI), SQLAlchemy and Alembic.
- Architected and shipped production AWS infrastructure supporting the platform's public API and frontend — ECS Fargate for backend services, RDS SQL Server for the database, S3 + CloudFront for the React SPA, and Bitbucket Pipelines for CI/CD.
- Developed an internal Python SDK wrapping hydrologic data analyses and water science computations into reusable OOP classes (using Pandas, NumPy, SciPy), reducing what previously required a full service's worth of logic to three lines of code, consumed natively in Python codebases as packages and exposed via APIs to Java and PHP applications across the organization.
- Built Airflow DAGs for sampling schedules, data analysis, and contamination result notifications across 2,000+ water systems; incorporated data mismatch resolution using LLM calls to automatically reconcile sample collection dates against lab reports, reducing manual correction effort by 90%.
- Implemented a checkout system using PayPal Expanded Checkout for eLearning course purchases, handling \$10,000+ in daily transactions, with CloudWatch for monitoring and alerts.

Software Engineer Intern - Graphwear Technologies | California, USA

Apr 2025 - Jul 2025

- Developed and deployed a FastAPI backend on AWS to ingest and analyze lactate measurements from a Bluetooth wearable, serving endpoints for the iOS app and internal dashboards; also contributed core iOS features in Swift including Bluetooth packet parsing every 30 seconds and observer-based state management for real-time lactate readings.
- Set up infrastructure-as-code and CI/CD pipelines with AWS CloudFormation and Bitbucket Pipelines, automating infrastructure provisioning, testing, and deployment end-to-end.

Software Engineer - Aaseya | Hyderabad, India

Aug 2020 – Jun 2022

- Worked on backend microservices in Spring Boot powering data exchange for a court management system with JWT authentication, centralized exception handling, and role-based access control.
- Implemented an end-to-end Hearings Management module in Pega, enabling judges to schedule, reschedule, and cancel hearings across multiple case types with REST APIs and role-based access for appellants, defendants, and judges.
- Contributed to workflows like case registration, travel ban and others on Pega using parent-child case hierarchies, circumstanced rules, cascading and authority matrix approvals; handled multi-stage processing using subprocesses, flows, and SLAs for time-bound escalations.

Software Engineer - JK Tech | Hyderabad, India

Jan 2019 – Aug 2020

- Developed backend REST APIs in Spring Boot and created multiple Pega modules for the Government of Andhra Pradesh's E-Pragati (OnePortal) platform, covering examination scheduling, real-estate registrations, Helpline Counter operations and integrations for CCAvenue payment gateway.

PROJECTS

Spotify AI Assistant (Flask & LangGraph)

- Built a conversational Spotify agent using a LangGraph ReAct loop, implementing OAuth 2.0, typed tool schemas, and multi-turn state management to expose Spotify operations — playlist creation, track discovery, recommendations, audio features and library management — as LLM-callable tools.

Formula1 Data Pipeline (Azure Databricks & Spark)

- Built a medallion ETL pipeline in Azure Databricks over a Delta Lakehouse with Unity Catalog to ingest Formula1 data into ADLS, applying PySpark transformations at each layer to produce curated, query-ready Delta tables for reports and dashboards.